
Integrated Forest Planning Across Multiple Ownerships and Decision Makers

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ABSTRACT. Social evaluation of public and private land-use planning of forests and rangelands is increasingly integrated across multiple land ownerships and resolved in a politicized forum of multiple decision makers. Multicriterial optimization models have limited utility for decision analysis in this environment as they must be formulated from the perspective of a single decision maker. A flexible integer programming model is demonstrated as a useful and appropriate tool to simulate and explore the collective options in a multiple ownership integrated planning (MOIP) problem. Multiple budget constraints are featured in this model as a vehicle by which resource contributions from both landowners and nonlandowners can be explicitly recognized. The hypothesis that MOIP problems define a positive sum game is supported by analyzing the data set of published national forest plan Draft Environmental Impact Statements (DEIS) for California data set. *FOR. SCI.* 37(1):200-226.

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INTEGRATED PLANNING ACROSS FOREST OWNERSHIPS is a multicriteria, multidecision maker problem not easily resolved using optimization analysis. A hypothesis, heretofore untested, is that the aggregate choices across landowners define a positive sum game with a net increase in social benefits, with winners capable of compensating the losers. Gains result from allowable cut type effects (Davis and Johnson 1986), and the consideration of new production resources and compensating institutional arrangements.

A framework for analyzing the Multiple Owner Integrated Planning (MOIP) problem builds on conventional planning for individual ownerships (or planning units) in a two-stage process. First, two or more feasible plans are developed for each planning unit to define choice options. Then aggregate MOIP choices (or decision variables) are defined by combining discrete plans for each planning unit. This paper documents the emergence, importance, and structure of the MOIP problem and suggests a flexible integer programming formulation for its analysis. The positive sum hypothesis is tested in a California case study.

EMERGENCE OF THE MOIP PROBLEM

National forest planning is clearly a MOIP problem. Diverse interests in the western states are realizing the importance of extensive federal land ownership to

their objectives. National forest plans frequently project long-term reductions in federal timber harvest, changes in wildlife habitat, altered visual quality, and important impacts on other amenities. The current round of USDA Forest Service planning has also, by design, encouraged more affected constituencies to participate and offer their opinion. The planning methodology currently employed by the Forest Service (FORPLAN models), has defined the collective production possibilities of our forests better than previous planning efforts. In most cases, unfortunately, it was discovered that the aggregate productive capacity of a forest is less than previously estimated by independently summing results from plans developed separately for each resource. The combined result is more people sitting at the national forest table trying to divide a smaller pie. The realities of this unpleasant situation are difficult for both the Forest Service and the national forest stakeholders to accept.

State governments, having been asked by the Forest Service and by their citizens to participate in the planning process, want to evaluate and comment on the aggregate and cumulative effects of changes in the management plans of the Forest Service within the context of outputs from all ownership sectors. Such analysis requires assemblage of comparable, comprehensive information about all wildlands and all owners within the state. Unfortunately each federal agency, state agency, and private landowner uses a different analytical system, data format, and timetable for data collecting and planning. As a result, critical information is not easily compared or arrayed to portray combined effects. Moreover most states have just begun to invest in obtaining and organizing the needed information.

Even for states willing and able to respond, it is unclear how federal plans should explicitly recognize and adjust for their comments. The Resources Planning Act (RPA) of 1974 says the Assessment shall "include a description of Forest Service programs . . . and the relationship of these programs and responsibilities to public and private activities." With regard to National Forest System Resource Planning, Section 5 of the RPA says plans developed for the National Forests shall be "coordinated with the land and resource management planning processes of State and local governments and other Federal agencies." The National Forest Management Act (NFMA) of 1976 adds no additional language over the RPA elaborating the relationship of Forest Service to state planning. The NFMA Regulations (1982), in section 217.9, define a "responsible line officer" to coordinate Forest Service planning with other planning efforts and a general procedure of review and discussion. These regulations fail to indicate how substantive conflicts identified by the states or other public agencies will be dealt with in the planning process, particularly when several forests are involved.

Exacerbating this problem is the Forest Service's treatment of each forest as an independent planning unit. Draft plans for the forests within a state are released over 10 to 20 months, each with a 90-day National Environmental Policy Act (NEPA) comment period constraint. Ideally, western states that need to process and evaluate up to 20 individual plans would like to be provided with all plans at about the same time and then be allowed a year or more for comment. This would permit the state to access and review federal data and analytical models, compile information on private lands and other public agencies, generate new options, conduct their own independent analysis, and prepare a thoughtful and comprehensive statewide response.

Currently several states are making a serious effort to overcome the con-

straints on their interactions with the Forest Service at a statewide level. Idaho requested and obtained a statewide analysis of the aggregate and cumulative impact of Forest Service plans on the state's economy and employment (La Vere 1986). The governor's office of Oregon, where timber supports some 50% of the state's economy, is assessing the combined impacts of projected federal and private timber supplies on the Oregon timber economy and developing a single coordinated state response supported by the state's congressional delegation. The detailed comments signed by Governor Goldschmidt on the Willamette National Forest DEIS (Goldschmidt 1988) and the draft, FORPLAN quantified, state alternative for the Siskiyou National Forest (Governors Forest Planning Team 1988) indicate the seriousness of Oregon's concern with federal planning. Oregon intends to provide such comment and a state alternative for every national forest in Oregon. California has just finished a comprehensive assessment of all private industrial and nonindustrial forest and rangelands within the state (California Dept. of Forestry and Fire Protection 1988). This information, when coupled with Forest Service DEIS data on the planning choices for 17 national forests, presents a comprehensive picture of what the future may hold for California if all plans are implemented. Serious questions have been raised about cumulative environmental, social, and economic impacts, and whether more integrated planning could produce greater net social benefits for California.

MODELS AVAILABLE FOR ANALYSIS OF MOIP PROBLEMS

A substantial literature, many problem formulation strategies, and several optimization techniques have emerged to deal with multiple objective problems since the early 1970s. Virtually all of this material addresses problems that have a single decision maker and two or more goals. These models and methods have been referred to as *multiobjective programming* (MOP), or more recently, *multicriteria decision making* (MCDM). Goal programming is the MCDM method that has probably received the most application in forestry (Bell 1976, Dyer et al. 1979, Arp and Lavigne 1982). For the most part these applications consider the allocation and use of public lands where the decision variables are acres allocated to different uses or cultural prescriptions. The decision maker is assumed capable of specifying initial goal targets and deviation weights for over or underachievement of these targets. Field et al. (1980) and Walker (1984) suggest that the best way to establish the initial goal targets is to determine the maximum attainment of each goal optimized separately over the constraints of the system. No consensus is evident on the best way to put weights on deviations within and between goals. Simulations with different weight sets are simply provided to the decision makers who are asked to resolve the problem. Steuer and Schuler (1978) used an interactive approach for a public land planning problem and clearly considered the forest planning team to be "the decision maker." In their study, the analysts used a filtering procedure to reduce the feasible region to a small set of representative choices. Then, the planning team was asked to select a preferred alternative from among this small set of choices at each sequential round. Mendoza et al. (1987) suggest that the small set of discrete planning alternatives presented does not fully span the feasible region and use MGA (Modeling to General Alternatives) methods to provide a more complete set of choices. Harrison and Rosenthal

(1986) developed a microcomputer multicriteria optimization procedure that asked landowners to reveal information on relative goal importance through pairwise preference comparisons of a small set of multiple output choices. Hotvedt et al. (1982) did a related study with heuristic ranking of harvest schedules by pulp and paper company foresters.

The MOIP problem, as defined earlier in this paper, implicitly or explicitly has multiple decision makers. Single decision maker (or decision team) models are not directly applicable. While the public has legislatively delegated administrative decision makers with full authority in public land management, they in reality find their role to be one of “referee” over a political decision process. MOIP problems will be “solved” only through political negotiation. Cohon and Marks (1975) recognize this important distinction in their review paper and comment that when political feasibility is a primary consideration, it requires “definition of the next problem in public planning: the multiobjective multiple-decision maker problem.” Steuer (1986) recently surveyed and organized the fast developing field of MCDM but offers few suggestions for application of multicriteria optimization techniques to the MOIP problem.

In short, the operations research literature has few direct suggestions for dealing with the multiple decision-maker multiple criterion problem. Single decision maker techniques have limited utility. Any “optimization” technique implies a consensus of all participants on goal levels, weights, and priorities. If such consensus is not forthcoming, optimization is inappropriate for analysis of the problem. Optimization techniques themselves, however, may have considerable value as information generation tools. By solving over a series of alternate goal and value structures, participants can be made much better informed of the shape of the feasible region (in decision and objective space) for themselves, and for other participating interests. This could help facilitate the negotiation and search process to find a consensus position.

Greenleaf and Harrison (1987) encountered such a problem when they worked with a Pennsylvania school board faced with the politicized problem of closing several elementary schools and reassigning students. A mixed integer model of the school system provided minimum cost solutions, but the board members and others manually manipulated the models to obtain more politically tractable solutions. The authors concluded “because the political process is an integral segment of decision making in this context, no model can be used for developing directly implementable solutions. However, quantitative models can be used to provide a starting point from which a workable solution can be derived.”

Robert Reich (1985) reviews the evolving social view of what a public administrator’s job is supposed to be. The pluralistic, interest group mediation view of the 1960s saw the public administrator as “a referee, a skillful practitioner of negotiation and compromise.” By the late 1970s net benefit maximization was the driving conceptual model for the public administrator who was “primarily an analyst, rather than referee.” Once distributional issues are assumed away with the Hicks-Kaldor criterion (Hicks 1939), the net benefit maximization tends to glorify big analytical models, big data sets, and optimization. Reich finds extreme interpretation of both paradigms unsatisfactory and argues the public administrator’s role is not to make administrative decisions but to help the public deliberate over the decisions that need to be made. Like Greenleaf and Harrison, he sees data, models, and analysis to have a vital but supporting role in a political process where

the players become technically educated about their options enroute to a politically feasible solution. Wondeleck's (1988) recent analysis of public participation in national forest planning supports this concept.

DEFINITION OF THE MOIP PROBLEM AS A DECISION PROBLEM

Within the geographic boundaries of the state, forests and other wildlands are distributed over thousands of different private and many different public owners. Most of these ownerships are a single planning unit, but the Forest Service and some large industrial companies are composed of several planning units. The many nonindustrial private ownerships within a geographic area can be aggregated to form planning "units" of all NIPF owners of a given behavior class. For each planning unit, at a given point in time none, one, or more than one mutually exclusive plan has been developed and written for the management of the unit. Even a unit without a documented plan is being managed in some way, and the results of this management can be estimated.

The decision variables for MOIP planning are built up from the known plans for each planning unit within a political or geographic region.

Let X_{k1} = plan 1 for planning unit k

$$\begin{aligned} X_{k1} &= \{0,1\} \\ k &= 1, \dots, K \\ 1 &= 1, \dots, L_1 \text{ for } k = 1 \\ &1, \dots, L_2 \text{ for } k = 2 \\ &\dots\dots\dots \\ &1, \dots, L_K \text{ for } k = K \end{aligned}$$

and, because plans are mutually exclusive,

$$\sum_1 X_{k1} = 1.0 \text{ for } k = 1, \dots, K.$$

This recognizes that plans are unique to each planning unit and that different numbers of plans are available for different units.

An aggregate choice associated with the region consists of one mutually exclusive plan for each planning unit within the region.

$$\begin{aligned} S_1 &= \text{regional aggregate choice} \\ &= [X_{1l} X_{2l} \dots X_{kl}] \end{aligned}$$

where $l = 1, \dots, L$.

The total number of MOIP choices will be the number of possible permutations of the mutually exclusive planning unit choices

$$L = L_1 L_2 \dots L_K.$$

A NUMERICAL EXAMPLE OF AGGREGATE CHOICES

To illustrate, a hypothetical data set describing outputs and costs of the choices for each planning unit is adapted from Davis (1988) (Table 1). The region of interest is a state that has four planning units.

TABLE 2.
Aggregate Planning Choices Available to the State.

Aggregate choice	Planning unit alternatives					Volume harvest (mmbf)	Area reserved (m ac)	Total cost (M\$)
S1	B1	I1	F1	N1		920	510	2400
S2	B1	I1	F2	N1		1020	610	2900
S3	B1	I1	F3	N1		820	610	2200
S4	B1	I1	F4	N1		720	610	2000
S5	B1	I1	F5	N1		620	910	1800
S6	B2	I1	F1	N1		840	610	2100
S7	B2	I1	F2	N1		940	710	2600
S8	B2	I1	F3	N1		740	710	1900
S9	B2	I1	F4	N1		640	710	1700
S10	B2	I1	F5	N1		540	1010	1500

FEASIBILITY OF AGGREGATE CHOICES

Initially, each individual choice for each planning unit is assumed feasible in that it can be implemented on the ground and that the projected outputs will be forthcoming. If each planning unit choice is feasible, then the aggregate choices are also feasible with respect to all planning unit constraints. An aggregate choice may not be feasible if, in addition, aggregate constraints exist. For example, there may be aggregate limits on federal budget, cumulative employment or sedimentation impacts. Political and social impacts that were not considered in developing choices for individual planning units may surface. If the aggregate budget is limited to

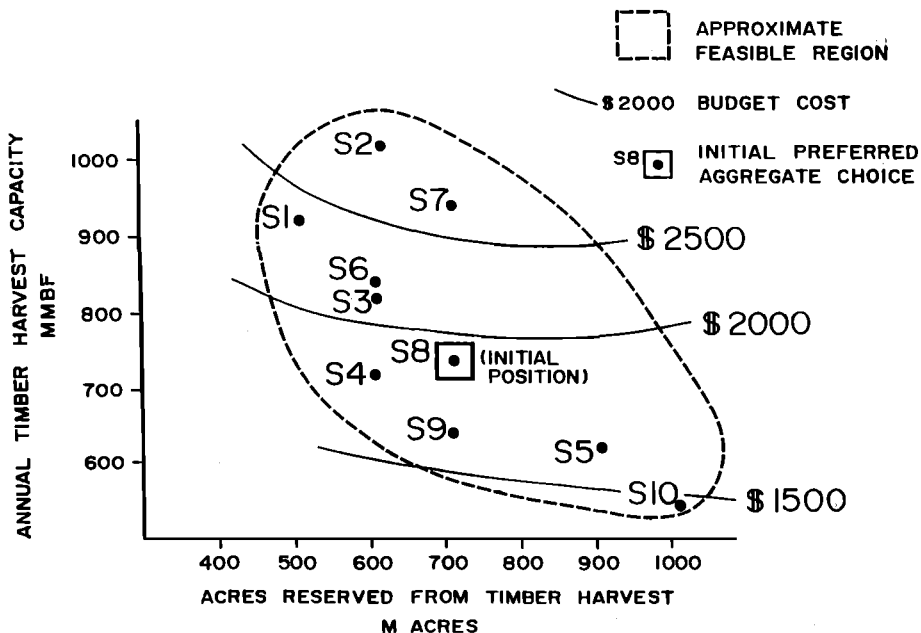


FIGURE 1. Discrete aggregate choices and approximate feasible region in decision space.

\$2000, S8 is the preferred choice in our example; the apparently “better” alternative S7 would be too costly.

Even if known aggregate constraints are satisfied, convex combination of existing solutions cannot be used to form new aggregate choices. Such combinations may not be feasible when disaggregated and the full range of spatial and other unique local constraints are imposed at the planning unit. Since the aggregate choice is made up of mutually exclusive, nondivisible unit plans, the aggregate choices are also mutually exclusive and nondivisible. Convex combinations at best provide suggestions for new choices to be developed on the planning units.

EVALUATION OF AGGREGATE CHOICES

The existence of multiple decision makers, even for individual planning units, is the essence of the MOIP problem. Each of these stakeholders brings a different value system and perspective to the problem. Two types of subjective approaches might be used by an individual stakeholder to identify a preferred choice (1) implicit subjectivity and (2) explicit subjectivity. For implicit subjectivity, the decision maker first defines a discrete (or continuous) representation of the feasible region in decision space. Then, instinctively accessing his or her utility function, the decision maker directly select a preferred solution from the feasible region in decision space. In explicit subjectivity, the decision maker attempts to articulate an explicit utility function. This utility function is used to calculate the value of each aggregate choice in decision space and, through optimization techniques, the “best” solution is found.

Through long experience using operations research technique for single decision maker problems under the dictates of economic theory, we have become comfortable with the conventional wisdom that methodologies that retain dominated alternatives for consideration are inferior. We usually implement this belief in a linear program by using a single set of value weights in an objective function to search the surface of a feasible region to find the optimal solution. We sometimes change the weights in the objective function [or the right-hand side (RHS) level of a goal related constraint] to locate efficient solutions from different points of view. As the literature and practice reveals us, value weights are always considered or assumed to be nonnegative and assigned numerical values on a 0.0 to infinity scale. See for example Haimes et al. (1975) or the statement by Kuhn and Tucker (1951). If the nonnegative assumption is true, then the dictate to consider only noninferior alternatives is correct. In fact, however, a good case can be made that some important forest outputs have negative utility to many persons and organizations. For example, consider the individual who detests the use of clearcutting in any form for reasons of esthetics, principle, or intrinsic value system. To this person, the amount of timber cut by clearcutting or the acres clearcut would be appraised using a negative weight, and the negative utility from clearcutting would be subtracted from the positive utility generated from other forest outputs. On the other hand, reserved old growth with high mortality or an unsalvaged wildfire might trigger negative utility to those who see this as needless “waste.” To some hikers, noisy off-road vehicles provoke negative responses. To some the idea or spectacle of game killing is negative, while to the traditional hunter and fish and game departments, it is strongly positive. The point is, nearly

every important forest output can invoke negative evaluations by some affected interest. The implication of this is that there is no unique set of efficient solutions over the decision makers for a MOIP problem. For public lands, all of these interests must be recognized as politically valid. Responding to the existence of negatively evaluated outputs in a MOIP problem context thus likely requires that we deliberately consider alternatives beyond the noninferior set identified for a particular decision maker. For multiple decision makers we need to examine the set of efficient solutions as perceived by all decision makers.

It's not clear whether the participants in negotiations over MOIP problems will want to use implicit or explicit approaches to evaluate choices and alternatives. Probably some of both. The answer to this will not be known until the people involved are actually assembled, and their analytical comfort levels are revealed. Each group of participants will surely approach the problem in a unique way.

The process of identifying participant goals, deciding the most important outputs for quantification, and assembling unit plan options into aggregate choices will constitute the principle analytical demand on an information system to support MOIP planning. We would expect most stakeholders to prefer making direct choices through implicit evaluation. In this case the analysis models will be used as simple simulators and accounting tools to add up the consequences of specified choices. If the participants wish to use more explicit evaluations, then they may desire to utilize a variety of search, generation, filtering, and selected optimization techniques of multicriteria decision making to better understand each other and the choices available.

For helpful analysis of MOIP problems we will likely have to abandon traditional insistence on working only with efficient solutions. In cases where negative utility or disutility is generated to some decision makers by some forest outputs, a given alternative solution that is considered dominated by one decision maker may be considered nondominated in the eyes of another. Likewise, the notion of optimality will be of little help to those who want to (or have to) subjectively select options. If pushed too hard, optimization may only serve to muddy communication and create mistrust between participants. Setting aside the powerful tool-concepts of efficiency and optimality seems wrenching and leaves little analytical elegance, but systems analysis and operations research still has an important job helping society deal with very important land management problems. The role of analysis will become more to define problems, educate political participants, and provide support information to facilitate negotiations.

A FLEXIBLE INTEGER PROGRAM FORMULATION FOR THE MOIP LAND PLANNING PROBLEM

Considering the need to relate to multiple decision makers, each with different analytical needs, an integer programming formulation was designed as a flexible information system to support iterative negotiation and learning.

Discrete plan choices are viewed as package deals, where any one alternative must be implemented completely or not at all. For example, National Forest planning unit alternatives are presented in a DEIS as mutually exclusive, and each can be regarded as (0,1) integer variable. Private land ownerships within a defined geographic area are typically classed as industrial or nonindustrial. If all owners in

each class are treated as a planning unit, alternatives for these private planning units are scenarios describing the distributions of how these lands will be owned and used in the future under existing and new policy packages. If scenarios are considered mutually exclusive, then these alternatives can also be treated as (0,1) integers. It might be desired, however, to let alternatives for the private planning split, realistically implying that part of the owners follow one scenario and the remainder other scenarios. If budgets to implement the policy packages are limited, then this would make sense. Accounting variables are defined to aggregate the production of outputs, activities, and consequences of interest from regional and statewide aggregations of the public and private forest planning units. "Optimal" combinations of plan alternatives can be identified as directed by the specific set of goals and constraints given for the problem.

Tradeoffs between conflicting criteria can be analyzed in several ways, including: (1) a weighted function of selected activities or outputs in the objective function, (2) mandatory output or activity requirements, (3) weighted deviations from desired levels of selected forest outputs, or (4) constraints that force pre-selected unit plan alternatives into solution. When goals are treated as targets and stated as equality constraints, deviation variables for over- and underachievement are explicitly defined. A traditional goal programming objective function can be established to minimize a weighted sum of these deviations. Alternative objective function concepts, such as maximizing a weighted function of the aggregate outputs or a minimax formulation, can also be used.

Such a flexible goal structure can accommodate the spectrum of objectives from present net worth (PNW) maximization to traditional goal programming to a pure simulation. Its structure in application will depend on how the people representing affected interests feel about articulating and negotiating their goals, and, on their perceived power within negotiation vs. their power in legal, political, and other means of achieving desired goals.

Budgets are the key device to describe the resources used by policy options and to explicitly portray the linkages between different institutions and landowners. Separate budget constraints can be defined within this model for the stage, the national forests, and any other owner class of interest such as industrial owners. New types of innovative linked options or joint planning alternatives can be defined that draw or require funding from more than one budget source and require implementation of specific alternatives by several planning units.

A generalized mathematical representation of this formulation of the MOIP problem is as follows:

Let

i = index of geographic region	$i = 1, 2, \dots, I$
j = index of owner type	$j = 1, 2, \dots, J$
k = index of planning unit	$k = 1, 2, \dots, K$
l = index of alternative plan for a planning unit	$l = 1, 2, \dots, L_1$ for $k = 1$ $1, 2, \dots, L_K$ for $k = K$

Then define

X_{ijkl} = the decision variable representing the implementation status of plan **l** on planning unit **k** of owner type **j** in region **i**.

Also

$$0 \leq X_{ijkl} \leq 1 \text{ for noninteger formulation}$$

$$X_{ijkl} = 0,1 \text{ for integer formulations}$$

$\sum_l X_{ijkl} = 1.0$ for mutual exclusivity of alternatives on a given planning unit (can also serve as a land constraint)

let

m = index of forest output type (these can be input or output activities)
 $m = 1, 2, \dots, M$

t = index of planning period $t = 1, 2, \dots, T$

r = index of budget source $r = 1, 2, \dots, R$

Then define

Q_{ijklmt} = the amount of output or consequence **m** provided in planning period **t** by implementing alternative **l** for planning unit **k** of owner type **j** in geographic region **i**.

Q_{imt} = the total amount of output or consequence **m** provided in planning period **t** for the plan options selected for subregion **i**.

$$= \sum_{jk} Q_{ijklmt}$$

Q_{jmt} = The total amount of output or consequence **m** provided in planning period **t** by owner type **j**.

$$= \sum_i Q_{ijklmt}$$

C_{jmt} = a relative value per unit associated with the production of output **m** in planning period **t** by owner class **j**.

G_{mt} = goal stating a target amount of output **m** desired in period **t**. Can be further specified by owner type **j** and region **i**.

Z^u_{mt} = Goal stating a mandatory maximum amount of output **m** required in period **t** (upper bound).

Z^l_{mt} = Goal stating a mandatory minimum amount of output **m** required in period **t** (lower bound). Upper and lower bounds can be further specified by owner type **j** and region **i**.

b_{ijkltr} = Budget cost of implementing plan **l** on planning unit **k** of owner type **j** in region **i** in period **t** from source **r**.

h_{tr} = Amount of budget from source **r** available in period **t**. Budgets can be further specified by unit **k** region **i** and owner **j**.

H_t = Total budget available in period **t**.

$$= \sum_r h_{tr}$$

also, define the deviation variables

d^+_{mt} = the amount by which a goal G_{mt} is overachieved

d^-_{mt} = the amount by which a goal G_{mt} is underachieved

and the relative value weights (positive or negative) assigned to the deviation variables.

w^+_{mt} = the relative importance per unit overachievement of goal G_{mt}

w^-_{mt} = the relative importance per unit underachievement of goal G_{mt}

An easily understood measurement of goal achievement is percent achievement of a goal target. A maximum and minimum value for a goal can be calculated by solving a linear program with this objective, subject only to mandatory constraints. Achievement is then scaled over the max to min range (1.0 = maximum achievement). Define the measures

P_{mt} = the proportion or amount by which goal m is achieved over the max-min range in period t

$$P_{mt} = (Q_{mt} - Q^{\min}_{mt}) / (Q^{\max}_{mt} - Q^{\min}_{mt})$$

Then define

α_t = the minimum achievement over two or more outputs in period t where
 $P_{mt} - \alpha_t \geq 0$ for all m used to define α_t

By maximizing α_t in the objective function, a classical MAXMIN solution is achieved.

A SUMMARY STATEMENT OF THE MODIFIED GOAL PROGRAM FOUNDATION

Alternative Objective Functions

$$\text{Minimize } \sum_{mt} [w^+_{mt} d^+_{mt} + w^-_{mt} d^-_{mt}] \quad (1a)$$

Minimize

$$\begin{aligned} & \sum_{mt} [w^+_{mt} (d^+_{mt} / Q^{\max}_{mt}) \\ & + w^-_{mt} (d^-_{mt} / Q^{\max}_{mt})] \end{aligned} \quad (1b)$$

$$\text{Maximize } \sum_{mt} (C_{mt} Q_{mt}) \quad (1c)$$

$$\text{Maximize } \alpha_t \quad (1d)$$

Constraints

$$\sum_{ijkl} Q_{ijklmt} X_{ijkl} - Q_{mt} = 0 \quad (2)$$

$$\sum_{ijkl} b_{ijklt} X_{ijkl} \leq H_t \quad (3a)$$

$$\sum_{ijklr} b_{ijklr} X_{ijkl} \leq H_{rt} \quad (3b)$$

$$\sum_l X_{ijkl} = 1.0 \quad (4)$$

$$X_{ijkl} = 1.0 \text{ for some } l \quad (5)$$

$$X_{ijkl} - X_{ijk'l} \leq 0 \quad (6a)$$

$$X_{ijkl} - X_{ijk'l} = 0 \quad (6b)$$

$$k = 1, \dots, K \quad k \neq k'$$

$$k' = 1, \dots, K$$

$$Q_{mt} - d^+_{mt} + d^-_{mt} = G_{mt} \quad (7)$$

$$Q_{mt} \geq Z^t_{mt} \quad (8a)$$

$$Q_{mt} \leq Z_{mt}^u \quad (8b)$$

$$Q_{ijmt} \geq Z_{ijmt}^l \quad (9a)$$

$$Q_{ijmt} \leq Z_{ijmt}^u \quad (9b)$$

$$Q_{mt} - [Q_{mt}^{\max} - Q_{mt}^{\min}]P_{mt} = Q_{mt}^{\min} \quad (10)$$

$$\alpha_t - P_{mt} \leq 0 \quad (11)$$

Equation (1a) is the conventional objective function for the archimedian goal programming. Equation (1b) is different from Equation (1a) in that the outputs are standardized to remove the arbitrary affects of measurement units. Equation (1c) maximizes the aggregate value of outputs across all owners over time. Equation (1d) is an objective that maximizes the minimum achievement over the goals of interest. Equation set 2 defines all the output and activity accounting variables of interest for reporting, constraining, and objective-setting purposes. Equation set 3 specifies the multiple budget constraints. Equation set 4 expresses the multiple-choice constraints for the ILP model. It also defines the area available in each planning unit. Equation set 5 is invoked when plan choices for some or all of the planning units are exogenously selected and hardwired into solution. Equation set 6 adds to the ILP model the contingency constraints when plan choices across two or more planning units are linked together as package deals. Equation set 7 defines the deviation variables relative to soft goal target levels for selected outputs and activities. Equation set 8a and 8b are the hard goal constraints setting minimum and maximum levels on selected outputs or activities. Equation set 9 is used if a lower (or upper) limit is also imposed on output from a specified set of owners and geographic regions. Equation set 10 defines the standardized achievement over the selected goals. Equation set 11 defines the minimum fractional achievement variable α enabling its use as an objective as shown in equation 1d.

EXAMPLE FORMULATION AND ANALYSIS OF A MOIP PROBLEM

The generalized equation set can be thought of as a menu from which specific application models can be constructed and modified. Certainly all equation types would not be used in the same application model. To illustrate its use for analysis of a MOIP problem, consider the example presented in Tables 1 and 2. Several new planning unit choices are first defined to enrich the example and to illustrate important points. These additions are presented as a dynamic scenario to capture the context in which multiple owner analysis functions.

A Scenario

After the initial assessment of the ten choices S1–S10, the stakeholders of the planning problem are revealed to be the four landowners, the state government, and the environmental coalition. These six meet to discuss individual and collective goals and look for ways to improve their aggregate performance. First, the participants decide that two additional forest outputs need to be quantified and tracked to represent their goals: (1) acres temporarily reserved from timber harvest (*TRA*) and (2) the volume of timber harvest in the first planning period (*H1*). The second result of this meeting is some additional new choices for all of

the planning units. These new choices are in two groups, independent choices and linked choices. Any, all, or none of the independent choices can be implemented. While if any one of the linked choices is selected, all other choices in the linked package must also be selected.

INDEPENDENT CHOICES

Alternative I2: The environmental coalition purchases an inventory easement from industry to hold 40,000 ac of old growth from cutting for the next 20 years.

Alternative F6: The governor's office, industry, and the environmental coalition successfully lobby a congressional delegation to increase the annual budget for target national forests by \$300,000. This would allow an increase of 50 mmbf in both long-term sustained yield (*LTSY*) and in first period harvest (*H1*) over that achieved by the preferred alternative *F3*.

Alternative N2: The state and industry split the \$200,000 cost of an expanded incentive program for nonindustrial owners. Presented with this subsidy, the nonindustrials are expected to increase their *LTSY* and *H1* by 50 mbf but lower their own investing by \$50,000.

LINKED CHOICES

Responding to unexpectedly heavy political pressure for better protection of critical wildlife habitat, the technical planning teams from all four ownerships meet with state representatives and the environmental coalition. Based on their earlier work to define *I2*, *F6*, and *N2*, they come up with a package of four new options, one for each planning unit. Politically it was essential that all four had to be implemented, or none could be used. The new linked or package choice is the set {*B3*, *I3*, *F7*, and *N3*}.

Alternative B3: With a boost of \$100,000 in budget and by preparing a joint timber plan with adjacent national forests, the BLM can move from their preferred plan (*B2*) and provide 6 mmbf of *LTSY*, maintain the permanent reserve area at 40,000 ac, and a first-period harvest of 5 mbf.

Alternative I3: The industry sells 10,000 ac of old growth outright to the state and places 30,000 ac in temporary reserve for a cost of \$200 million to the state and \$200 million to the environmental coalition. Industry spends an additional \$100 million to intensify management on its reduced land base and produces a first period harvest of 20 mbf and a *LTSY* of 18 mbf.

Alternative F7: In addition to the activities under plan *F6*, those national forests which have large areas of mature sawtimber agree to a 50 mbf timber harvest departure over the next 2 decades, on the condition that annual monitoring shows industry, the BLM, and the NIPF are, in fact, effectively investing in timber production on their own lands at the agreed upon rates. The state contributes \$50,000 worth of state personnel time to help prepare and administer the accelerated Forest Service timber harvest program.

Alternative N3: Nonindustrial owners adjoining the 10,000 ac parcel of old growth sold by industry to the environmental coalition agree to sell an additional 10,000 ac to complete the old growth state park at a cost of \$50,000 to the state. Industry spends an additional \$50,000 in the NIPF incentive program to hold a *LTSY* of 23 million and a first-period harvest of 20 on the reduced NIPF land base.

THE EXAMPLE FORMULATED AS A LINEAR PROGRAM

The original data set of Table 1, as augmented by the new planning unit choices is shown in Table 3 for budget costs and Table 4 for outputs. The augmented

problem is shown in Figure 2 as a detached coefficient matrix for the linear programming formulation. General formulation Equations (1) through (11) should be easily tracked in the matrix. The matrix illustrates a structure that could be used to specify a family of problems.

Columns 1–16 define the primary decision variables for this example, which are the discrete planning unit choices. The first four rows tally the four primary outputs of interest, *LTSY*, *PRA*, *TRA*, and *H1*, and report their level for any solution through accounting variables defined for this purpose in columns 17–20. Constraining aggregate outputs or defining objective functions on aggregate outputs will make use of these accounting variables.

Rows 5–9 show the budget expenditures required from the five different budget sources in order to implement any of the planning unit choices. Budget expenditures could also be formulated as accounting variables. Note the financial involvement of the nonlandowning stakeholders in the augmented choices. The -0.25 entries for the state in row 6, columns 12 and 13, are the \$25,000 profit anticipated from the park leasing operation of *F6* and *F7*. The Forest Service lease income shows up as the \$10,000 reduction of federal budgets for *F6* and *F7* from 11 to 10.9 in row 5. In this model formulation the budget rows would play an important communication role in public deliberation by explaining just who pays for what benefits. They also represent the idea of multiple resource inputs that could be expanded to include multiple source facilities, labor, and equipment.

Rows 10–13 are the area constraints that say the land resource of each planning unit can only be used once.

Rows 14–18 link the package choice option $\{B3, I3, F7, N3\}$ to make it an all-or-nothing deal. These four variables must also be specified in the *LP* execution as 0,1 integer variables to complete the choice. If the package is to be selected, setting any one variable to 1 (such as shown in row 18) will force the package to be chosen.

TABLE 3.
Budget Costs of Augmented Planning Unit Choices.

Planning unit	Unit choice	Federal	State	Industry	Environmental	Nonindustrial
BLM	B1	5				
	B2	2				
	B3	3				
Industry	I1			6		
	I2			6	.5	
	I3		2	7	2	
Forest Service	F1	10				
	F2	15				
	F3	8				
	F4	6				
	F5	6				
	F6	10.9	-2.5			
	F7	10.9	-2.5	.5		
Nonindustrial	N1					3
	N2		1	1		2.5
	N3		1.5	1.5	.5	2

PLANNING UNIT:			BLM			INDUSTRY			FOREST SERVICE							NIPF					FOREST OUTPUTS					GOAL ACHIEVEMENT					SIGN	RHS		
UNIT CHOICES:			B1	B2	B3	I1	I2	I3	F1	F2	F3	F4	F5	F6	F7				N1	N2	N3	LTSY	PRA	TRA	H1	a	PLTSY	PPRA	PTRA	PHI				
COLUMN NUMBER:			1	2	3	4	5	6	7	8	9	10	11	12	13				14	15	16	17	18	19	20	21	22	23	24	25		26		
PRIMARY OUTPUTS			12	4	6	20	20	18	40	50	30	20	10	35	35				20	25	23	-1										=	0	
(1) Long term sustained yield (LTSY)																																=	0	
(2) Acres permanently reserved from final harvest (PRA)			30	40	40	1	1	2	10	20	20	20	50	21	21				10	10	11	-1										=	0	
(3) Acres temporarily reserved from final harvest (TRA)						40	30													10		-1										=	0	
(4) 1st period harvest (H1)			8	2	5	25	23	20	40	50	30	20	10	35	40				15	20	20			-1								=	0	
BUDGET COSTS			5	2	3				10	15	8	6	6	10.9	10.9				1	1.5													FB	
(5) Federal (FB)								2						-.25	-.25				1														SB	
(6) State (SB)						6	6	7							.5				1	1.5													IB	
(7) Industry (IB)								.5																									EB	
(8) Environmental (EB)																		3	2.5	2												NB		
(9) NIPF (NB)																																		
AREA CONSTANTS			1	1	1																													
(10) BLM						1	1	1																									1.0	
(11) Industry									1	1	1	1	1	1	1																		1.0	
(12) Forest Service																																	1.0	
(13) NIPF																			1	1	1												1.0	
LINKED ALTERNATIVES																																		
(14) B3, I3						1																												0
(15) E, F1								-1							-1																		0	
(16) B3, F7								1							1																		0	
(17) B3, I3						-1																											0	
(18) (if forced) I3								1																									0	
DIRECTLY SELECTED ALTERNATIVES																																		1
(19) All preferred B2			1																															1
(20) Forced I1						1																												1
(21) F3																																		1
(22) N1																			1															1
PERCENT GOAL ACHIEVEMENT																																		
(23) LTSY																																		-49
(24) PRA																																		-51
(25) TRA																																		0
(26) H1																																		-47
MAX-MIN GOAL																																		
(27) LTSY																																		0
(28) PRA																																		0
(29) H1																																		0

Figure 2. Detached coefficient matrix for example MOIP problem formulated as an integer programming problem.

TABLE 4.
Augmented Planning Unit Choices.

Planning unit	Unit choice	Long-term sustained yield (LTSY)	Acres permanently reserved from final harvest (PRA)	Acres temporarily reserved from final harvest (TRA)	First period timber harvest (HI)
BLM	B1	12	30		8
	B2	4	40		2
	B3	6	40		5
Industry	I1	20	1	40	25
	I2	20	1	30	23
	I3	18	2		20
Forest Service	F1	40	10		40
	F2	50	20		50
	F3	30	20		30
	F4	20	20		20
	F5	10	50		10
	F6	35	21		35
	F7	35	21		40
Nonindustrial	N1	20	10		15
	N2	25	10		20
	N3	23	11	10	20

Rows 19–22 represent the direct approach to hardwiring solutions, in this case the independently preferred alternatives *B2*, *I1*, *F3*, and *N1*. It's likely that this would be the most frequently chosen approach by a nontechnical political negotiators, intent on exploring the implications of different options and wanting to be in complete control of the simulation.

Rows 23–26 are calculations for reporting the fractional achievement of each goal over its maximum–minimum range. Using the calculated values of $LTSY^{\min} = 49$ and $LTSY^{\max} = 107$, the standardized achievement equation for *LTSY* is

$$-LTSY + 58PLTSY = -49$$

as shown in row 23 of Figure 2. Similarly, the standardized achievement equations for *PRA*, *TRA*, and *H1* are shown in Rows 24–26.

These max–min ranges can be normatively narrowed by the decision makers to form different membership functions for each goal and the fractional achievement variables of columns 22–25 used in a variety of objective functions, for reporting, and for communication between stakeholders.

One appealing use of the fractional achievement variables is to formulate a MAXMIN problem where α is defined as the minimum fractional achievement over several different goals. In the example formulation *PLTSY*, *PRA* and *PH1* were used to constrain α in rows 27–29. To use α , the objective function would be defined as maximize α , which would solve the problem to do as well as possible with the most difficult to achieve goal.

The example problem was solved with four different objectives and highlights how different interests might interpret the problem.

Problem run	Forest output				Maximum goal α	Variables in solution
	<i>LTSY</i>	<i>PRA</i>	<i>TRA</i>	<i>H1</i>		
All preferred choices	74	71	0	72	0.357	<i>B2, I1, F3, N1</i>
All independent choices	84	72	40	80	0.375	<i>B2, I2, F6, N2</i>
All linked choices	79	78	40	85	0.482	<i>B3, I3, F7, N3</i>
Max α over <i>LTSY, PRA, H1</i>	85.4	86.2	10	86.4	0.628	<i>B3, I1, 0.66F2, 0.34F5, N3</i>
Max α ; 0, 1 Integer over <i>LTSY, PRA, H1</i>	79	78	40	80	0.482	<i>I3, N3, F6, B3</i>

We see that the all independent and all linked runs provide more of every output and a correspondingly higher value of α . When α is maximized over 3 of the 4 goals (omitting *TRA*) in a noninteger formulation, then the decision variables split, and the amount of the omitted goal is reduced significantly, and α is numerically higher. The last run forces all variables to be integers and the same solution as the linked solution is obtained. The goal program to minimize underachievement produced yet another solution. What this means is that different decision makers will interpret the world differently and analysts need to provide flexible tools to provide the range of results illustrated here.

CALIFORNIA CASE APPLICATION

The 17 national forests in California constitute Region 5 of the USDA Forest Service. These forests were grouped into the six geographic subregions used by the California Department of Forestry and Fire Protection (1988) for its comprehensive assessment of California's natural resource systems (Table 5). Full anal-

TABLE 5.
California national forests and geographic subregions used for testing of study hypothesis.

Subregion	National forest
North Coast	Six River
Northern Interior	Klamath
	Lassen
	Modoc
	Shasta Trinity
Sacramento	Mendocino
	Plumas
	Tahoe
	Eldorado
San Joaquin	Inyo
	Sequoia
	Sierra
	Stanislaus
Southern California	Cleveland
	San Bernardino
Central Coast	Angeles
	Los Padres

ysis of California as a MOIP problem awaits development of comparable alternatives for the industrial, nonindustrial, and other public owners of the state. However, the 17 national forests were treated as independent planning units in order to test the research hypothesis that a positive-sum gain is possible through aggregate planning. The primary analysis was statewide, but a smaller model was also defined that considered only the four forests of the Northern Interior sub-region.

The model described by Equations (1)–(11) was quantified for the national forest system of California using output and activity data for alternatives published in the draft EIS of each forest for planning under the NFMA. Each forest had quantified and presented from 6 to 11 discrete plans or choices and these were treated as 0,1 integer variables in the mixed integer linear programming models. The six subregions were explicitly recognized, and the 51 different forest activities and outputs that were quantified in all 17 DEIS are tracked in the model over the first 5 decades of the plans. Of these outputs, only dispersed recreation and wilderness, developed recreation, anadromous fish, water yield, grazing, and timber production were priced in the objective function. The resulting LP model contained 942 rows, 287 columns, 127 integer variables. All runs were made using the MPSIII program on the IBM computer.

Preliminary analysis of this data (Davis 1988) showed dispersed recreation and wilderness, anadromous fish, and water to be relatively constant across all plans for virtually every forest. Except for a few forests, the level of the budget (operating cost) was not used as a constraint in forest planning. Cost, however, varied substantially across forest plans and the aggregate choices. Additionally, the amount of area designated for retention visual quality objectives varied significantly across the choice set and interacted strongly with timber production, budget and all other forest outputs.

Based on this preliminary work, four outputs (developed recreation, grazing, timber production, and retention area) and one input, operating cost, were used to see if a positive-sum gain could be found. A series of runs were made maximizing and minimizing each output independently for the statewide and Northern Interior models. These results, along with those from hardwired runs forcing the preferred alternative for each forest into solution, are shown in Table 6.

Recreation, grazing, timber, and budget are for the first decadal planning period, and retention area for the fifth decade. Detailed examination of the solutions showed that differences in first period output were proportionately maintained in subsequent time periods within 1%. Because of this, first period results are a reasonable index of the dynamic pattern of outputs and inputs. Virtually all outputs and inputs trended up with time.

Hypothesis Test

The hypotheses is that by considering aggregate planning choices, positive gains in forest outputs can be obtained from a given land base, budget, and other input resources, and positive gains in forest outputs can be obtained relative to those obtained by independent planning and choices. The test version of this hypothesis used for this study is “the four outputs of developed recreation, grazing, timber production, and retention area can be provided at aggregate levels greater than the amount provided by the preferred alternatives, while using equal or less

TABLE 6.

Preferred, maximum, and minimum possible amounts of primary forest outputs from national forests in California.

Primary output	Unit	Period	All PRF	Maximum	Minimum
<i>Statewide</i>					
Developed recreation	M RVD	1st	13972	14324	11599
Grazing	M AUM	1st	600	720	500
Retention	M Acre	5th	2967	5116	1532
Timber	MMBF	1st	1848	4523	3617
Operations cost	MM \$	1st	345	401	248
<i>Northern Interior</i>					
Developed recreation	M RVD	1st	1623	—	—
Grazing	M AUM	1st	217	—	—
Retention	M Acre	5th	625	—	—
Timber	MMBF	1st	630	—	—
Operations cost	MM \$	1st	95	—	—

operating cost than the preferred alternatives." To search the feasible region of the aggregate choice set for solutions that demonstrated this, a series of different runs and objective functions were used.

Run PRF\$MIN. The objective was to minimize operating cost, subject to the constraints that all outputs had to be provided at levels greater than or equal to preferred plan output levels. If the outputs could be provided by a set of plan choices at lower than the preferred costs, then the hypothesis is supported.

Runs P1–P4. The objective was to minimize the unweighted sum of negative (underachievement) deviations from output targets defined as the preferred level of outputs. The runs were made using budget constraints that varied in 25% steps across the min to max range of the operating cost. If solutions could be found that equalled or exceeded preferred outputs at costs equal or lower than preferred costs, then this supports the hypothesis.

Runs P9–P12. Repeat runs P1–P4 for the Northern Interior region.

Runs M1–M4. The objective was to minimize the unweighted sum of negative deviations from output targets defined as the maximum independent level of each output. $\text{Min } \sum_j (d^-_j)$ is the formal specification.

Runs M5–M8. The objective was to minimize the unweighted sum of the negative deviations $\sum_j (d^-_j)'$ from maximum output targets, where $(d^-_j)' = d^-_j / Q^{\text{max}}_j$.

Runs M9–M12. Repeat runs M1–M4 for the Northern Interior region.

RESULTS AND DISCUSSION

The first period output and cost levels for the runs are summarized in Table 7 and the output level, as a percentage of preferred output level, is given in Table 8. A graph of runs P1–P4 expressed as percent of preferred is given in Figure 3 and

TABLE 7.

Amount of primary forest outputs and operations cost generated from cumulative impact analysis in region five.

Scenario	Developed recreation (M RVD)	Primary grazing (M AUM)	Output retention (M ac)	Timber (mmbf)	Operations cost (MM \$)	Budget constraints
All PRF	13972	600.5	2967	1848	344.70	
PRF1\$MIN	14037	614.6	3315	1855	308.11	
<i>Statewide</i>						
P1	14051	570.5	3517	1665	286.41	≤286.5
P2	14069	614.3	3440	1850	321.31	≤324.8
P3	14018	618.8	3495	1899	344.61	≤363.2
P4	14018	618.8	3495	1899	344.61	≤401.5
M1	13844	521.6	4890	1401	286.26	≤286.5
M2	13852	560.1	4956	1624	324.61	≤324.8
M3	13852	564.2	4956	1713	330.81	≤363.2
M4	13852	564.2	4956	1713	330.81	≤401.5
M5	13759	568.2	4745	1368	286.40	≤286.5
M6	13767	568.8	4929	1630	323.11	≤324.8
M7	13767	572.8	4929	1719	329.31	≤363.2
M8	13767	572.8	4929	1719	329.31	≤401.5
<i>Northern Interior</i>						
All PRF	1623	217.1	625	630	95.1	
P10	1623	191.4	607	629	73.5	
P11	1623	228.0	658	695	100.2	
P12	1623	228.0	658	695	100.2	
M10	1623	189.8	869	471	73.5	
M11	1623	191.5	1157	607	89.2	
M12	1623	191.5	1157	607	89.2	

for runs M1–M4 in Figure 4. Again, differences in first period outputs were found to accurately reflect future output differences

Run PRF\$MIN shows that all preferred output levels can be exceeded at a cost that is 10.6 percent (\$36 million) less than the preferred aggregate operating cost. Runs P2, P3, and P4 also found solutions that provided up to 0.33, 3.5, 17.5, and 2.7 percent more of the four outputs using budgets that were less than that of the preferred. Runs P11 and P12 showed the same sort of gains for the Northern Interior region. These results support the hypothesis of this study.

Comparing runs P1–P4, P9–P12 to runs M1–M4, M9–M12 reveals a very different distribution pattern in aggregate outputs between solutions to the two target goal levels. Minimizing negative (underachievement) deviations from the achievable targets produced solutions that were well balanced across the four outputs. Minimization of negative deviations from the unobtainable maximum targets, however, produced solutions where one output (retention acres) was provided about 65% in excess of preferred levels, while the other outputs were provided some 5 to 12% less than preferred levels. When the deviations from the maximum goals were normalized as in runs M5–M9, similar results were obtained. The solutions are obviously sensitive to the method of goal setting and

TABLE 8.

Outputs and operations cost generated from the cumulative impact analysis in region five as the percentages of those from the ALLPRF scenario.

Scenario	Developed recreation	Grazing	Retention	Timber	Cost
PRF\$MIN	100.46	102.34	111.73	100.38	89.38
P1	100.57	95.00	118.54	90.10	83.09
P2	100.69	102.30	115.94	100.11	93.21
P3	100.33	103.05	117.80	102.76	99.97
P4	100.33	103.05	117.80	102.76	99.97
M1	99.08	86.86	164.81	75.81	83.05
M2	99.14	93.29	167.04	87.88	94.17
M3	99.14	93.96	167.04	92.69	95.97
M4	99.14	93.96	167.04	92.69	95.97
M5	98.48	94.62	159.93	74.03	83.09
M6	98.53	94.72	166.13	88.20	93.74
M7	98.53	95.39	166.13	93.02	95.54
M8	98.53	95.39	166.13	93.02	95.54
P10	100.0	88.0	97.0	99.0	77.0
P11	100.0	105.0	105.0	110.0	105.0
P12	100.0	105.0	105.0	110.0	105.0
M10	100.0	87.0	139.0	74.0	77.0
M11	100.0	88.0	185.0	96.0	93.0
M12	100.0	88.0	185.0	96.0	93.0

deviation weighting. Some good reasons for a particular set of weights and targets would be needed to recommend a solution as having real-world meaning.

Budgets were strongly constraining at the lower <\$286.5 million level (runs P1, M1, M5, P9, and M9), and had a moderate effect at the <\$324.8 million level (runs P2, M2, M6, P10, and M10). The 1987 Region 5 budget is about \$300 million, so the preferred set of alternatives and most of the combinations produced in this study would be infeasible without budget increases.

The aggregate results mask a great deal of changes in the plans selected by the models for individual forests. The plan chosen for the statewide runs of this study are given for each of the 17 forests in Table 9 and for the Northern Interior regional runs in Table 10. Each forest used a different set of abbreviations and codes for their alternatives but generally the ones that say CUR, RPA, or MKT emphasize commodities and those labeled AMN or LBU stress amenities. The Forest Service's Preferred plan (PRF) is seldom selected for any forest. As we would expect, at each budget level the forests are sorted out by their relative advantage at producing the different outputs; forests that can produce timber efficiently select a plan emphasizing timber, and forests that produce retention acres efficiently select plans with large areas in retention. For example, of the timber forests, the El Dorado and the Plumas consistently tend to produce timber while the Sequoia and the Lassen tend to provide amenities. The real reasons for this can only be found in the data underlying the numbers provided in each forest's DEIS. Again, results of the runs with maximum targets are somewhat different

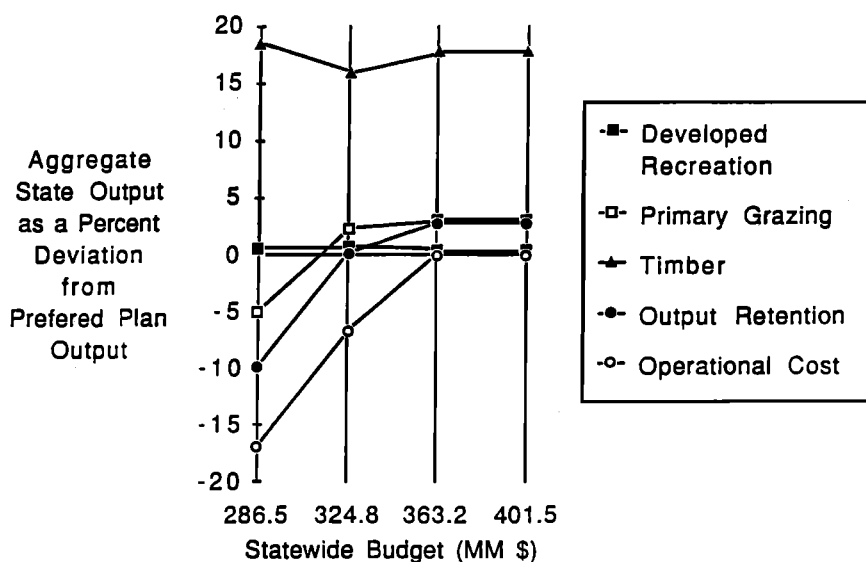


FIGURE 3. Outputs and operating cost as a percent of amounts provided by selecting preferred plans for all R-5 national forests at four budget constraint levels when plans selected minimizing underachievement of all preferred target levels (Runs P1-P4).

than the results of the runs with preferred output targets. More forests maintain amenity type plans at all budget levels.

These results only suggest some possibilities and questions that Region 5 might profitably address. It might make some sense for different forests to specialize more in outputs where they have a relative efficiency. The strong tradeoff be-

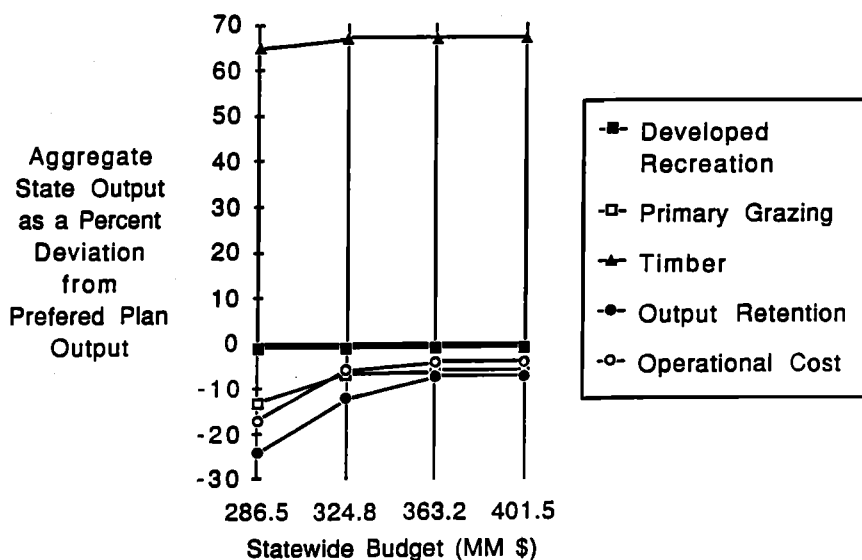


FIGURE 4. Outputs and operating cost as a percent of amounts provided by all preferred plans for all R-5 national forests at four budget constraint levels when plans selected minimized underachievement of maximum attainable levels of outputs (Runs M1-M4).

TABLE 9.
National forest plan option selected for statewide cumulative analysis.

Planning unit	All PRF	Scenario				Number different	Scenario				Number different	
		P1	P2	P3	P4		M1	M2	M3	M4		
Statewide												
Angeles	PRF	PRO4	PRO4	PRO4	PRO4	PRO4	0	RPA3	PRO4	PRO4	PRO4	2
Cleveland	PRF	CUR2	AMN5	AMN5	AMN5	AMN5	2	CUR2	PRF1	PRF1	PRF1	2
El Dorado	PRF	CUR2	MKT5	MKT5	MKT5	MKT5	2	CUR2	PRF1	PRF1	PRF1	2
Inyo	PRF	CEE4	CEE4	RPA3	RPA3	RPA3	2	PRF1	PRF1	PRF1	PRF1	1
Klamath	PRF	AMN5	AMN4	ABC1	ABC1	ABC1	2	AMN5	ABC1	MKT7	MKT7	3
Lassen	PRF	AMN5	AMN5	COM6	COM6	COM6	2	AMN5	AMN5	AMN5	AMN5	1
Los Padres	PRF	NOW6	NOW6	NOW6	NOW6	NOW6	0	NOW6	NOW6	NOW6	NOW6	1
Mendocino	PRF	PRF1	RPA3	MKT4	MKT4	MKT4	3	LBU6	AMN5	AMN5	AMN5	2
Modoc	PRF	RBU5	IND4	CUR2	CUR2	CUR2	3	AMN6	AMN6	AMN6	AMN6	1
Plumas	PRF	PRF1	PRF1	PRF1	PRF1	PRF1	0	PRF1	PRF1	PRF1	PRF1	1
San Bernardino	PRF	CUR2	CUR2	CUR2	CUR2	CUR2	0	CUR2	CUR2	CUR2	CUR2	1
Sequoia	PRF	CUR2	PRO8	AMN4	AMN4	AMN4	3	AMN4	AMN4	AMN4	AMN4	1
Shasta Trinity	PRF	CUR2	CUR2	MIX7	MIX7	MIX7	2	MIX7	MIX7	MIX7	MIX7	1
Six River	PRF	DFG8	RPA3	RPA3	RPA3	RPA3	2	NMK5	LOW7	LOW7	LOW7	2
Sierra	PRF	LBU4	LBU4	MKT7	MKT7	MKT7	2	LBU4	PRO8	PRO8	PRO8	2
Stanislaus	PRF	D	A	G	G	G	3	F	A	A	A	2
Tahoe	PRF	AMNE	AMNE	MRCF	MRCF	MRCF	2	AMNE	AMNE	AMNE	AMNE	1

TABLE 10.

National forest plan options selected for regional cumulative analysis of California's Northern Interior.

Planning unit	Scenario						
	All PRF	P10	P11	P12	M10	M11	M12
Klamath	PRF	AMN5	MKT7	MKT7	AMN5	MKT7	MKT7
Lassen	PRF	AMN5	AMN5	AMN5	AMN5	AMN5	AMN5
Modoc	PRF	RBU5	IND4	IND4	RBU5	AMN6	AMN6
Shasta Trinity	PRF	CUR2	PRF1	RF1	MIX7	MIX7	MIX7

tween timber and retention area, which is a reasonably good proxy for the many commodity vs. amenity tradeoff issues, suggests that a detailed analysis of the empirical and spatial character of the tradeoff might be informative.

Because many outputs and activities were not included in the deviation and cost minimization objective functions, their changes did not affect the selection of plans for the different runs, and their opportunity costs are not measured in this study. In addition, there are obvious deviations in the spatial distributions of outputs relative to the all-preferred set of solutions. This implies significant changes in the incidence of costs and benefits to forest users. For example, changes in spatial distribution of use and possible increased transportation costs were not recognized or evaluated in this analysis. The political, social, and legal acceptability of these changes is unknown. For these reasons, the results of this study do not and cannot support a conclusion that the preferred alternatives are not the set that provides the greatest net social benefit.

CONCLUSIONS

Multiple ownership, multiple decision maker integrated planning problems are emerging as the essential land management problem confronting most public and many private landowners. MOIP problems appear in three situations: (1) a single owner with one planning unit and multiple decision makers such as one national or state forest; (2) one owner, multiple planning units, and multiple decision makers such as all of the forests in a Forest Service Region; and (3) multiple owners, multiple planning units, and multiple decision makers such as planning across all landowners within the state. Land use planning by publicly held corporations can also fit these categories. For all of these cases, traditional single decision maker multicriteria optimization or decision models have limited application for resolution of these essentially political problems. Logic, hypothetical examples, and a California case study all support the hypotheses of MOIP that aggregate net social gains can be achieved through development and consideration of aggregate planning choices across multiple ownerships. Capturing this gain will require planning forums where informed, trusting stakeholders understand the full scope of feasible aggregate choices and acceptable ways can be found for the winners to compensate the losers of a negotiated solution.

A flexible integer programming formulation allowing alternative direct control

by MOIP problem stakeholders, provides a useful organizing framework for information supporting public deliberation and political negotiation.

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